Polynomial Regression Curve Fitting Challenge

Machine Learning Assignment 6

October 7, 2026

1 Introduction

Given a dataset with a clearly non-linear relationship, we fit Polynomial Regression models to understand how varying the polynomial degree affects the resulting fit. Specifically, we compare a polynomial of degree 2 versus a polynomial of degree 4 to visually build intuition regarding the concept of overfitting.

2 Visualization

The plot below demonstrates the differences in how each polynomial degree models the scattered, noisy data.

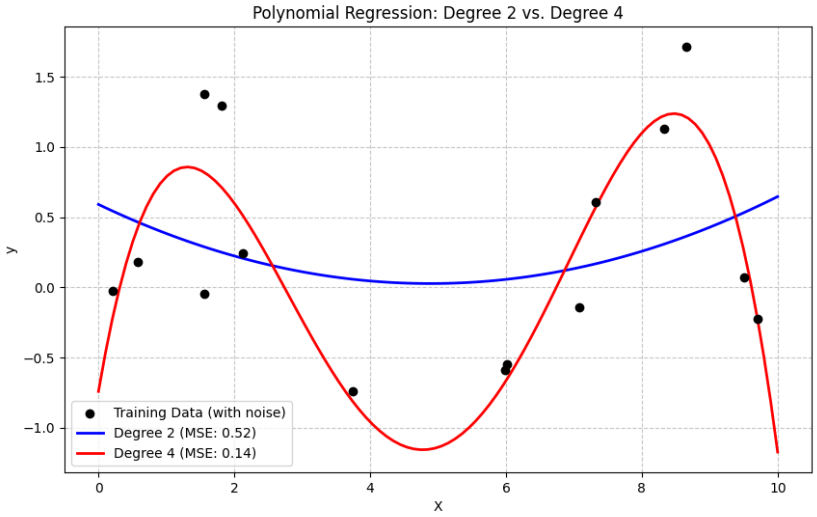


Figure 1: Comparison of Polynomial Degree 2 and Degree 4 fits.

3 Explanation: Overfitting in Polynomial Regression

In this challenge, we compare a Degree 2 polynomial (quadratic) with a Degree 4 polynomial to examine their behavior on the training data.

3.1 Degree 2 (Blue Line)

This model captures the general downward and upward trend of the data but might be slightly too simple (underfitting) to capture the true underlying sine wave perfectly. However, it remains smooth and generalizes reasonably well to the overall shape of the dataset without reacting to minor noise.

3.2 Degree 4 (Red Line)

The Degree 4 model has much more flexibility due to its higher number of parameters. As seen in the plot, it tries much harder to pass exactly through the scattered training points. This results in a curve that wiggles sharply, especially at the edges or between points.

While the Mean Squared Error (MSE) on the training data is typically lower for the Degree 4 model, this is a classic sign of **overfitting**. The model is capturing the random noise in the training set rather than the true underlying pattern. If we introduced new, unseen test data, the Degree 4 model would likely produce massive errors due to its extreme fluctuations, whereas the smoother Degree 2 model would generalize much better.